

Digits

## Error Log Entry

### **What error message did you encounter (if any)?**

There was no error message during the execution of the final version of the program. The calculations needed correction while creating the program.

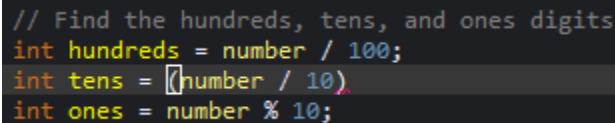
### **What unexpected behavior did your program exhibit?**

Initially, I encountered difficulties making the tens digit correct. The code had to perform the right division and % operation to split all three digits.

### **What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)**

The issue was a logic error because I was not using the correct calculation for the tens digit. I needed to use `(number / 10) % 10` to get the tens place.

### **Include a screenshot of specific lines of code.**

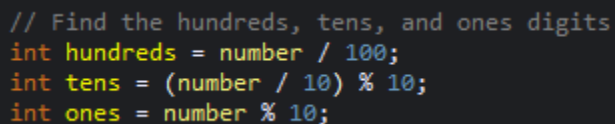


```
// Find the hundreds, tens, and ones digits
int hundreds = number / 100;
int tens = (number / 10);
int ones = number % 10;
```

### **How did you fix the issue?**

I fixed the issue by changing the tens calculation to `(number / 10) % 10`. I then tested the program with different three-digit numbers to make sure all the digits were correct.

### **Provide the corrected code or solution using a screenshot.**



```
// Find the hundreds, tens, and ones digits
int hundreds = number / 100;
int tens = (number / 10) % 10;
int ones = number % 10;
```

Change

## Error Log Entry

### What error message did you encounter (if any)?

I did not get an error message in the final version of my program. I had to make sure my calculations were correct so the program would give the minimum number of coins.

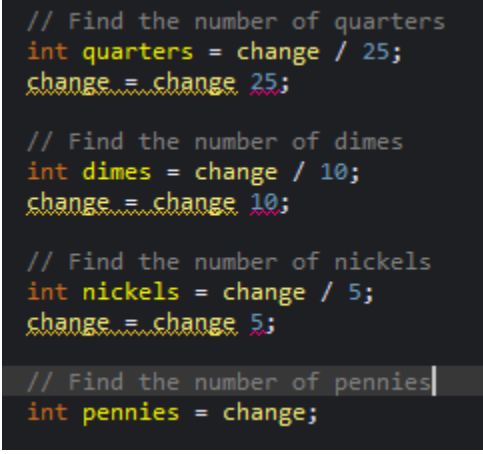
### What unexpected behavior did your program exhibit?

I found it hard to manage the amount of leftover change after computing each type of coin at first because the program was required to compute the next coin using the leftover change.

### What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

The issue was also a logic error because I needed to update the amount of change after finding each type of coin. I used the % operator to find the remaining change after quarters, dimes, and nickels.

### Include a screenshot of specific lines of code.

A screenshot of a code editor showing C++ code for calculating the minimum number of coins. The code is as follows:

```
// Find the number of quarters
int quarters = change / 25;
change = change % 25;

// Find the number of dimes
int dimes = change / 10;
change = change % 10;

// Find the number of nickels
int nickels = change / 5;
change = change % 5;

// Find the number of pennies
int pennies = change;
```

### How did you fix the issue?

I fixed the issue by updating the % **variable** after calculating each type of coin. This allowed the program to use only the remaining change for the next coin.

### Provide the corrected code or solution using a screenshot.

```
// Find the number of quarters
int quarters = change / 25;
change = change % 25;

// Find the number of dimes
int dimes = change / 10;
change = change % 10;

// Find the number of nickels
int nickels = change / 5;
change = change % 5;

// Find the number of pennies
int pennies = change;
```